

Phase-2

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Date of Submission: [Insert Date]

Github Repository Link: [Update the project source code to your Github Repository]

1.Problem Statement

Type of Problem: Classification (predicting AQI category like Good, Moderate, Unhealthy) OR **Regression** (predicting exact AQI value).

Helps authorities take preventive actions.

Crucial for public health advisories.

Supports environmental policy planning.

Project Objectives

Goal: Build a machine learning model that predicts the air quality index (AQI) accurately.

Technical Objectives:

- o Clean and prepare the data for modeling.
- o Perform EDA to uncover insights.
- o Engineer new meaningful features.
- o Train and compare multiple machine learning models.
- o Evaluate models based on accuracy (for classification) or RMSE (for regression).

3. Flowchart of the Project Workflow

Start -> Data Collection -> Data Preprocessing -> EDA -> Feature Engineering -> Model Building -> Model Evaluation -> Results Interpretation -> End

I can create a visual flowchart for you too if you want.

4. Data Description

- o **Dataset Source:** [Mention your source, e.g., UCI Machine Learning Repository, Kaggle].
- o **Type:** Structured (tabular) data.
- o **Records and Features:** [You can mention the number after checking the dataset].
- o **Static or Dynamic:** Static.
- o **Target Variable:** AQI Level or AQI Value.

5. Data Preprocessing

- o Handle missing values (mean/median imputation or removal).
- o Remove duplicate records.
- o Detect and treat outliers.
- o Encode categorical variables (One-Hot Encoding or Label Encoding).
- o Normalize or Standardize continuous variables if needed.

6. Exploratory Data Analysis (EDA)

o Univariate Analysis:

- o Histograms for each feature.
- o Boxplots for outlier detection.

o Bivariate/Multivariate Analysis:

- o Correlation heatmap.
- o Scatterplots: Feature vs Target.

o Insights:

- o Identify strong predictors (e.g., PM2.5, O3, NO2).
- o Understand distribution across AQI categories.

7. Feature Engineering

- o Creating new features (e.g., average pollution level, pollution/weather ratios).
- o Binning AQI values into categories if needed.
- o Feature selection using correlation or feature importance.

8. Model Building

- o Models to use:
 - o **Random Forest** (strong for tabular data).
 - o **XGBoost** (excellent for structured data and competition-winning).
- o Additional options:
 - o Decision Tree, Logistic Regression (if classification).
- o **Evaluation:**
 - o Classification: Accuracy, F1-Score, Precision, Recall.
 - o Regression: RMSE, MAE, R^2 Score.

9. Visualization of Results & Model Insights

- o Confusion matrix (if classification).
- o ROC-AUC curve (classification performance).
- o Feature Importance graph.
- o Residual plot (if regression).

10. Tools and Technologies Used

- o **Programming Language:** Python
- o **IDE:** Jupyter Notebook / Google Colab
- o **Libraries:**
 - o pandas , numpy for data handling
 - o seaborn,matplotlib,plotly for visualization.
 - o scikit-learn ,xgboost for machine learning.

11. Team Members and Contributions

Harini.R

Role: Data Preprocessing & Exploratory Data Analysis (EDA)

Contributions:

Collected and cleaned the dataset (handling missing values, outliers, encoding).

Performed univariate and multivariate data analysis.

Identified key trends and potential feature correlations.

Rani Blessy.S

Role: Feature Engineering & Model Development

Contributions:

Created and refined features based on EDA insights.

Built and trained multiple machine learning models (e.g., Random Forest, XGBoost).

Tuned hyperparameters and improved model performance.

Aspriya.M

Role: Model Evaluation, Visualization & Reporting

Contributions:

Evaluated model performance using accuracy, precision, recall, and F1-score.

Generated visualizations (confusion matrix, ROC curves, feature importance).

Compiled documentation and maintained the GitHub repository.